

Hybrid III 6 y/o Lab

BME254L - Spring 2026 - Palmeri

Dr. Mark Palmeri, M.D., Ph.D.

2026-01-07

Table of contents

Tasks	1
What to submit	2
Bottom Plate	4
Top Plate	6
Nodding Plate	8
Molded Neck	10
Lower Neck Bracket	13
Load Cell	14
Assembly	15

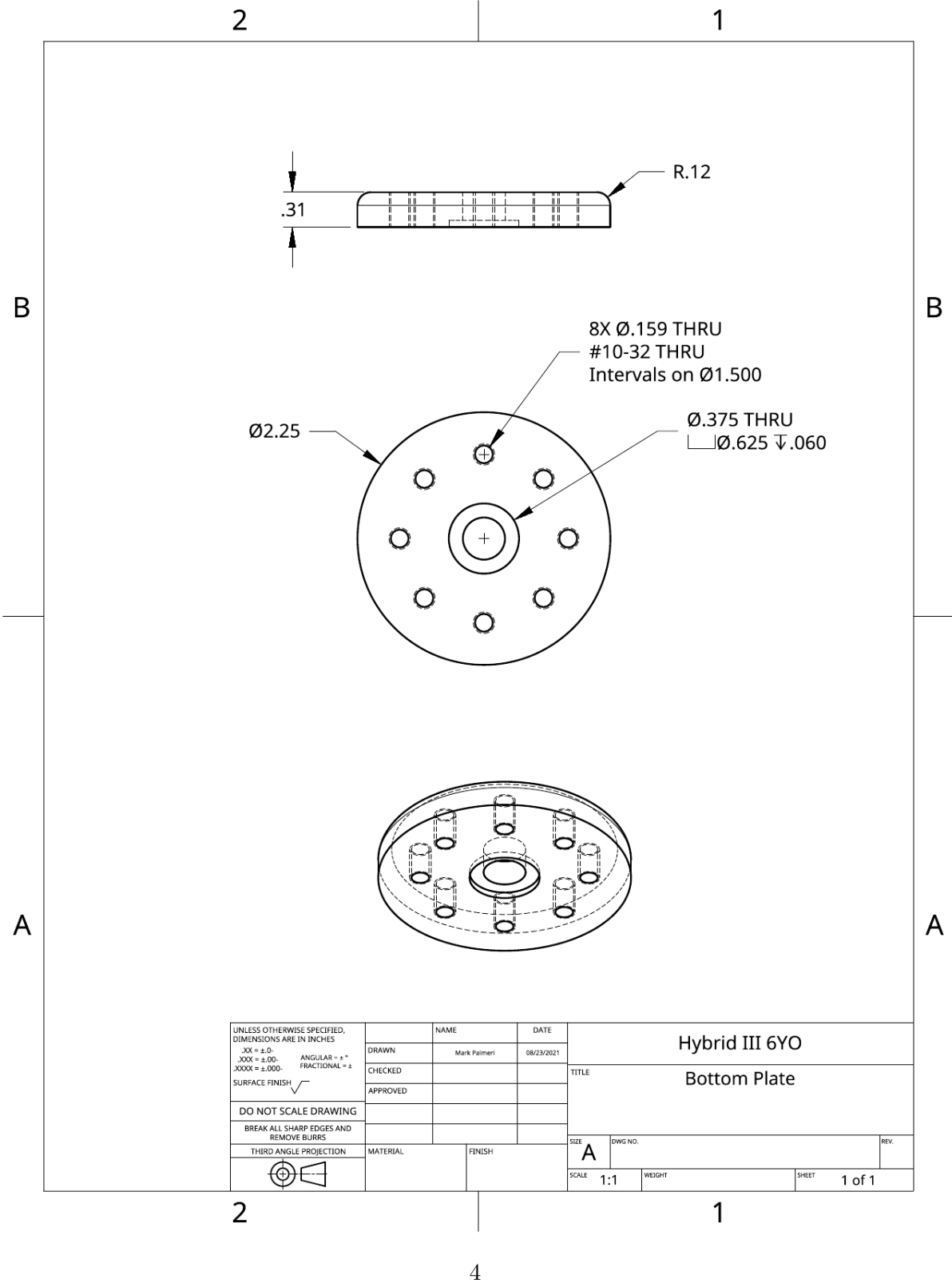
Tasks

- Create all of the parts for the Hybrid III 6yo test setup based on the attached drawings.
- You should create all of these parts and their associated drawings and assemblies as part of the same “Document” and share that document with Dr. Palmeri and the TA(s).
- Create the mechanical drawings associated with each of those parts (i.e., recreate the drawings that you are using to generate the parts) and the assembly.
- Assemble your parts as guided by the assembly diagram and as shown in the uploaded animation.
- There are online tutorials on Assemblies if you need some refreshers:
 - <https://learn.onshape.com/courses/introduction-to-assembly-design>
 - <https://learn.onshape.com/courses/fundamentals-onshape-assemblies>

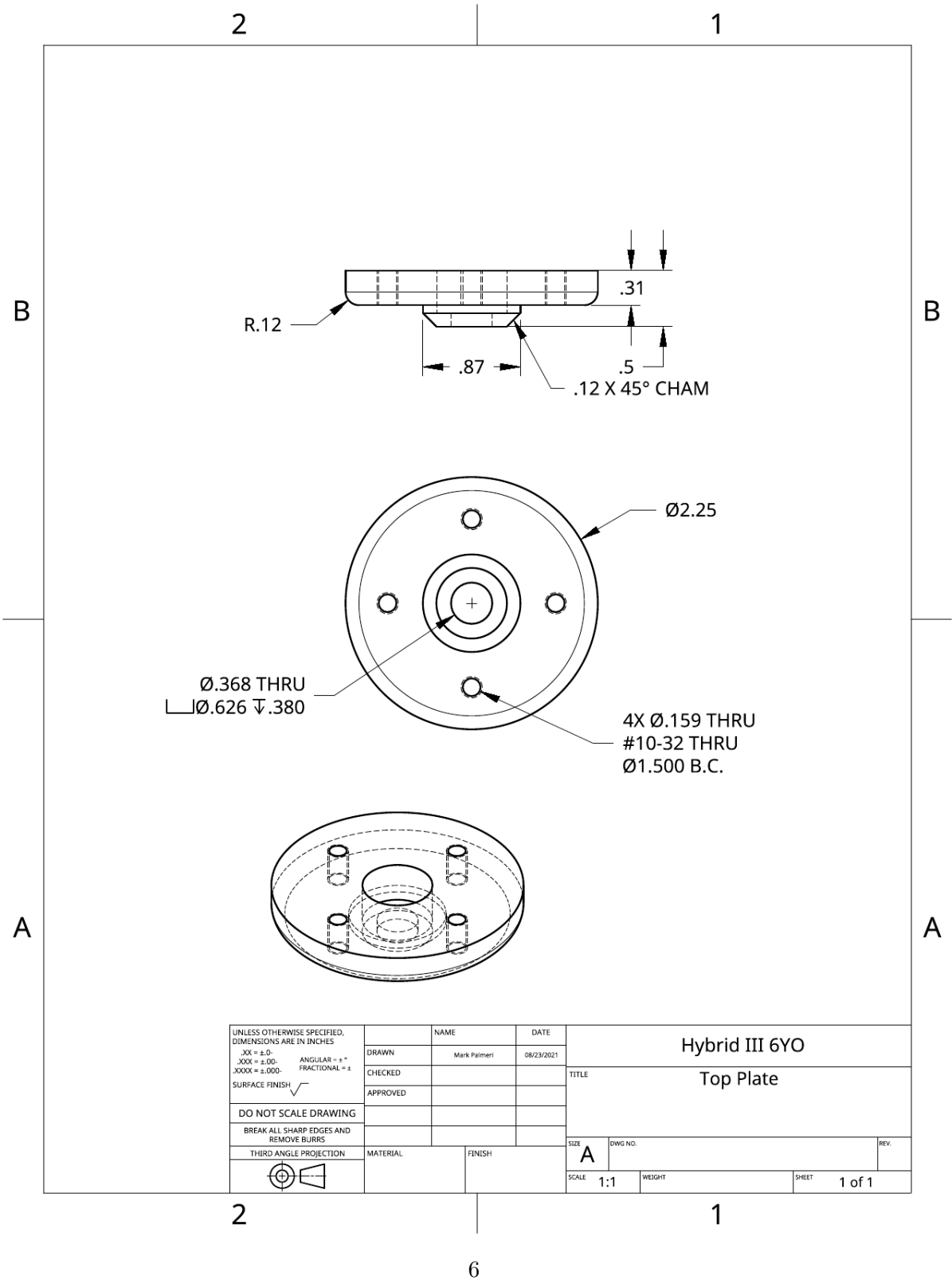
What to submit

Upload mechanical drawings and a screenshot of the part for all of your parts to the Hybrid III 6 y/o Gradescope assignment.

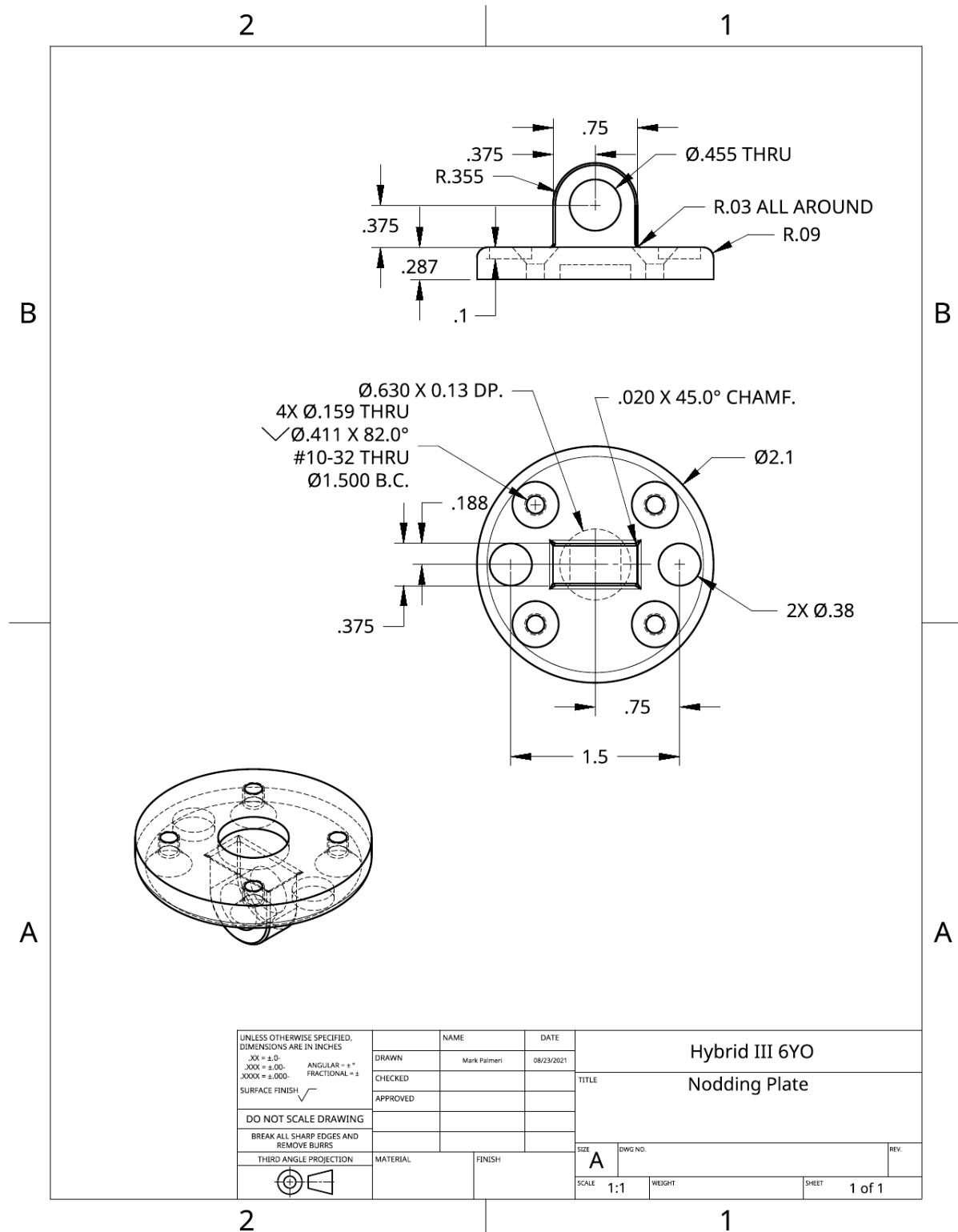
Bottom Plate



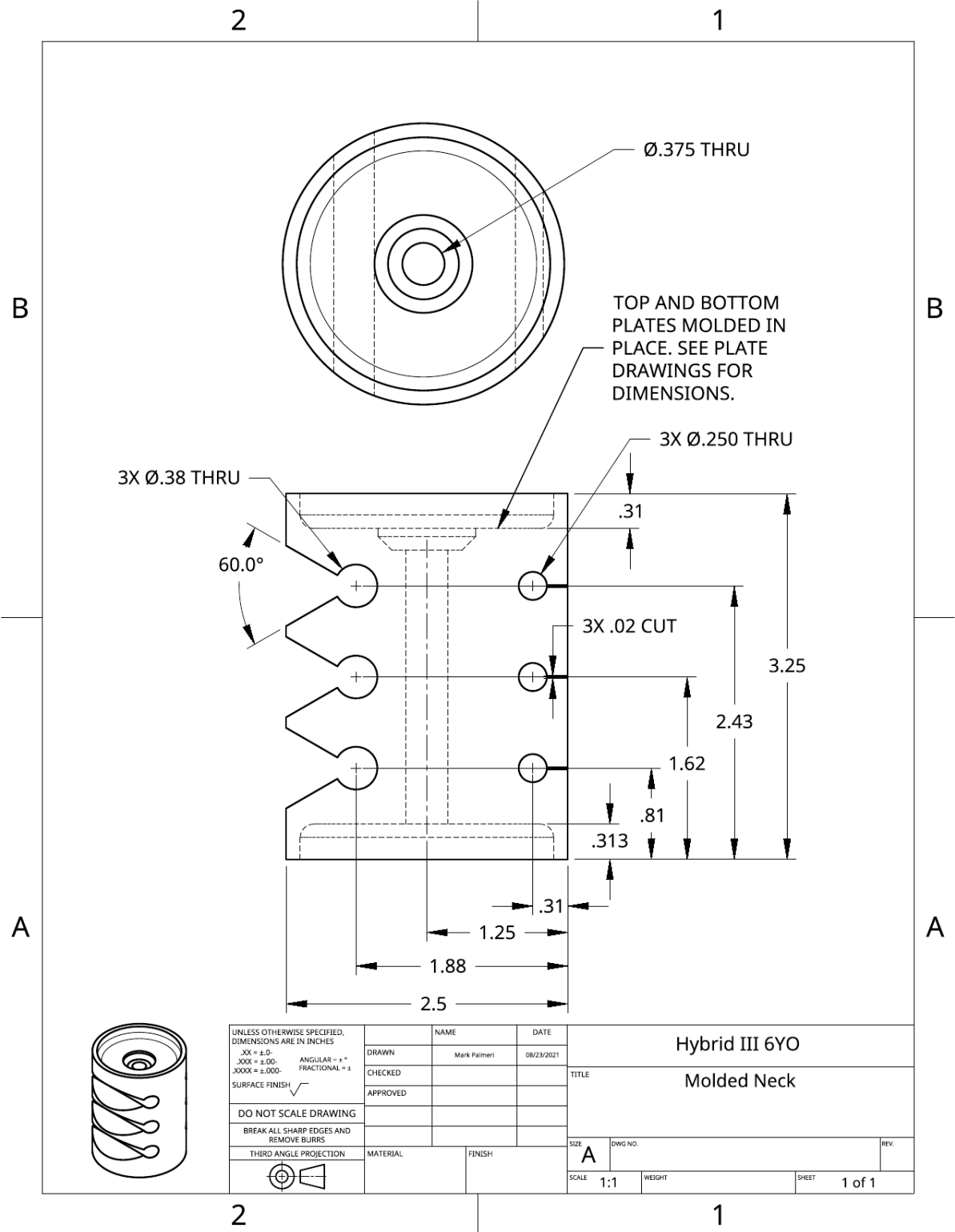
Top Plate



Nodding Plate



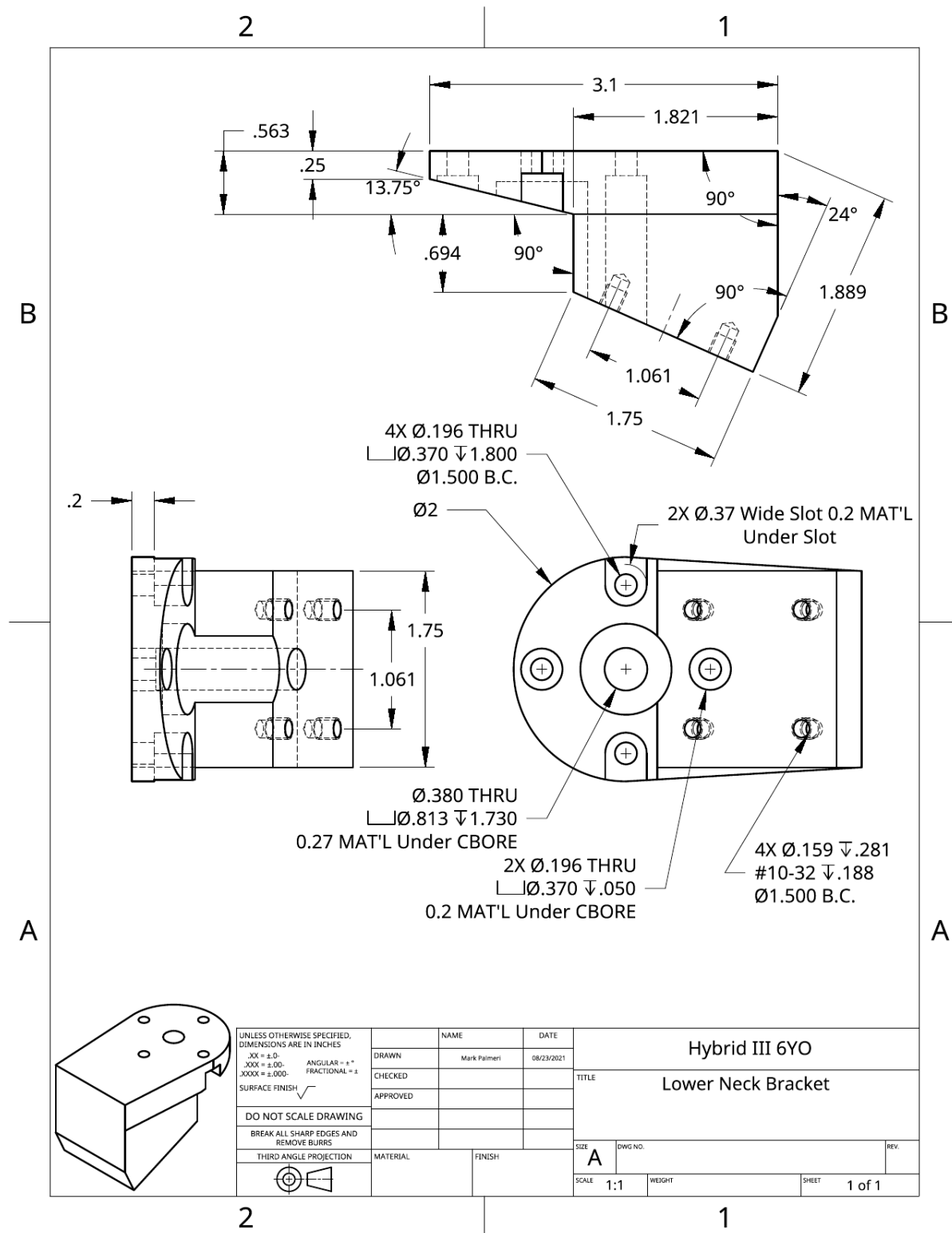
Molded Neck



 Tip

Note that the this part has features that are dependent on the top and bottom plates. To reduce the overhead of manually entering these dimensions from the other parts, you can use the **Derived** tool in Onshape to bring in the relevant faces from the other parts. <https://cad.onshape.com/help/Content/derived.htm#StepsDesktop>

Lower Neck Bracket



4 3 2 1

D

C

B

A

2.25
.75
.37
2
1
.5
R.25
4X R.25
.25
.5

90°
4X 25°
Ø.315
4.75°
1.25
1.05
1.6
1.26
1.08
.92
.58
.24
1.8
2.4
Ø.68
.72
.32
.17
.108

1.06
.57
.25
.12
Ø.438 X 0.040
R1.779
4X Ø.20
1/4-20
.700
1.5
1.471
.438
2.12
1.14
1.5

2X Ø.125
0.25 DP.
4X Ø.150 THRU
#10-24 X .125
X 0.25 DEEP TAP

4 3 2 1

Hybrid II

Load

UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES		NAME	DATE
XXX = ±.01	ANGULAR = °	DRIVEN	08/23/2021
XXX = ±.05	FRACTIONAL = 1/16	CHECKED	
XXX = ±.000		APPROVED	
SURFACE FINISH			
DO NOT SCALE DRAWING			
BREAK ALL SHARP EDGES AND REMOVE BURRS			
THIRD ANGLE PROJECTION			
MATERIAL		FINISH	

SCALE 1:1

WEIGHT

Assembly

